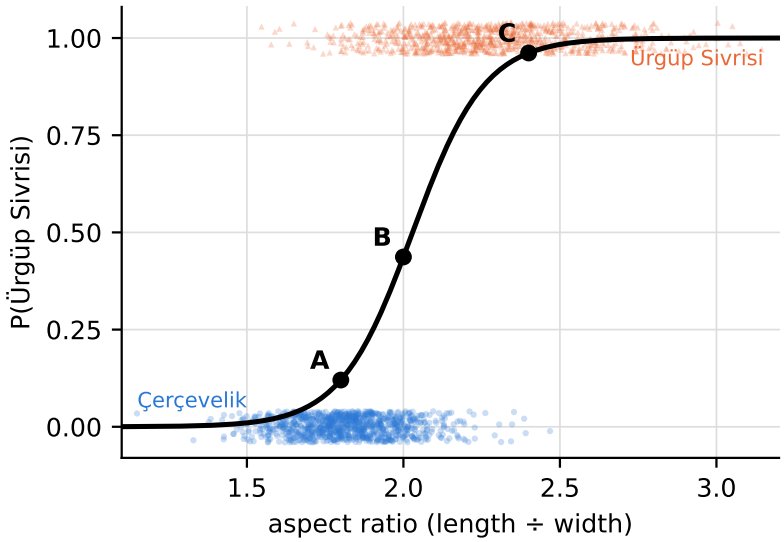
DATA 202 — Logistic Regression · Week 11 Monday

Pumpkin seeds of two varieties, Çerçevelik and Ürgüp Sivrisi, measured from photos. Your professor will tell you when to do each part.

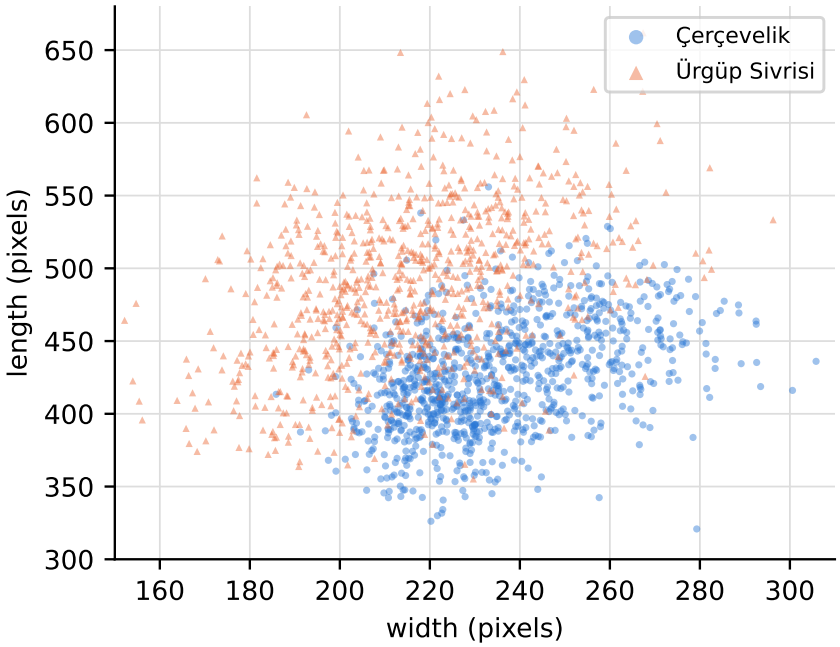
1 · The S-curve



Each dot is a training seed: Ürgüp Sivrisi on top, Çerçevelik at the bottom. The curve is the model's probability that a seed is Ürgüp Sivrisi.

1. **Mark** the 50/50 point on the curve.
2. **Shade** the aspect ratios where the model predicts Ürgüp Sivrisi.
3. Predicted variety of seed
   A: _____
   B: _____
   C: _____
4. **Circle** the seed the model is least sure about.

2 · Draw the boundary

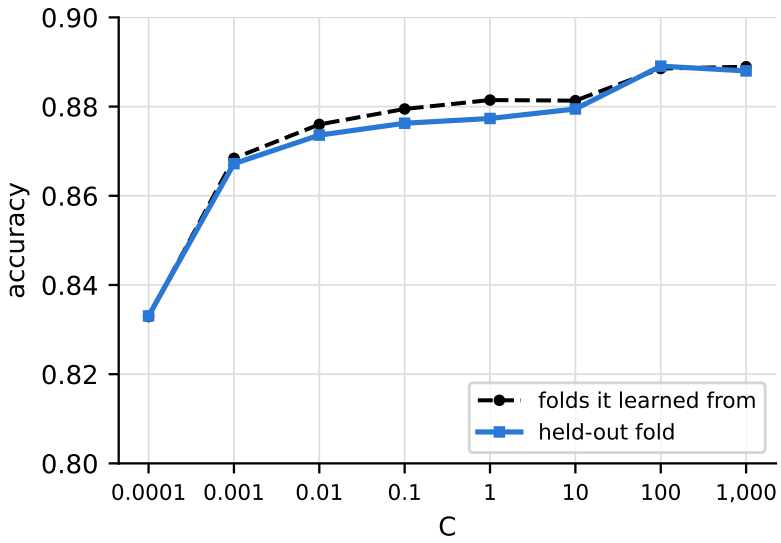


Training seeds by length and width.

1. **Draw** the straight line you would use to separate the two varieties.
2. **Circle** a region where no straight line could get every seed right.
3. Your line says "long for its width." Which column of the data measures that?

   _____

3 · Choose C



Logistic regression on all 12 columns (scaled), 5-fold cross-validation on the training seeds.

1. **Circle** the best C.
2. **Label** the underfitting side.
3. Is there an overfitting side? _____ **Pair talk:** why not?
4. Here a *larger* C means a _____ model. For Ridge, a *larger* alpha meant a _____ model.